

ECO102: Introduction to Macroeconomics

Lecture 6: Economic Growth

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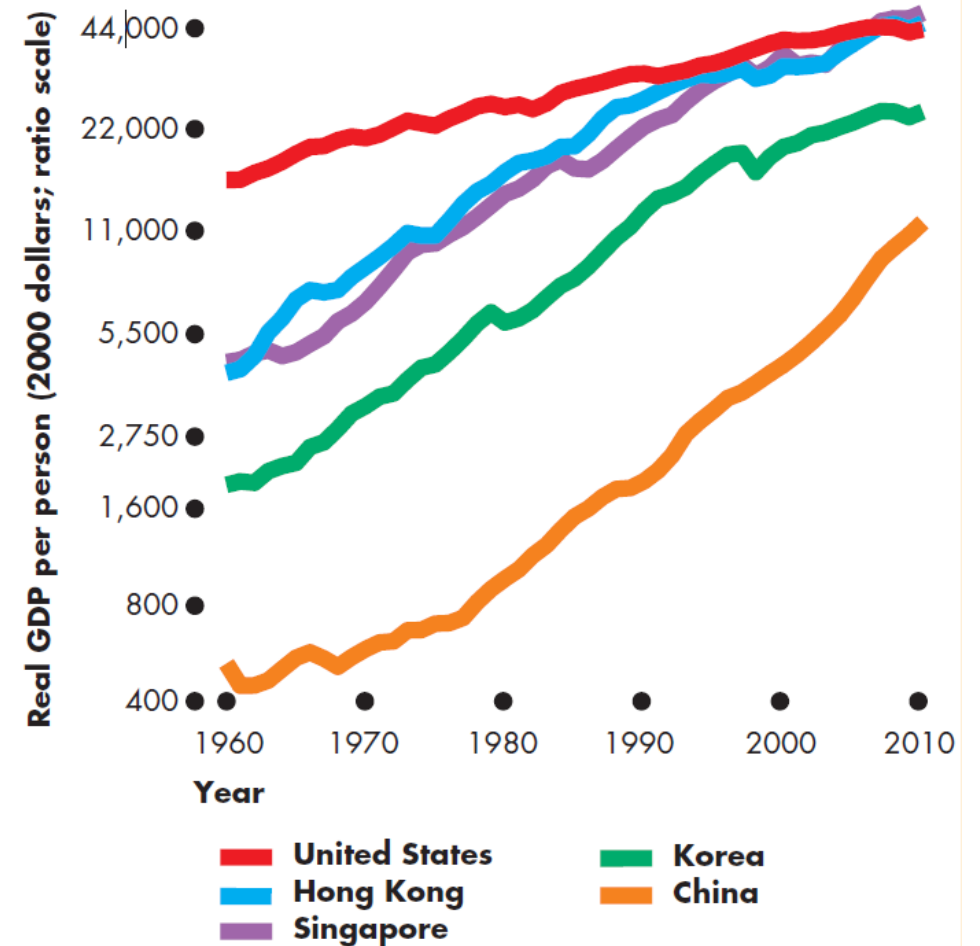
Rule of 70

- The Rule of 70 allows us to estimate how many years it might take for an amount to double if it is compounding each year
- If we divide 70 by the annual percentage growth rate of the variable, we get the number of years it would take for that variable to double
- Real GDP per person doubles in 70 years (70 divided by 1) if the growth rate is 1 percent a year. It doubles in 35 years if the growth rate is 2 percent a year and in just 10 years if the growth rate is 7 percent a year.

Economic Growth Trends

- Growth Trends allows us to see the standard of living across countries
- The East Asian Countries went through rapid expansion in the late 20th Century
- Between 1976 and 2010, Hong Kong transformed itself from a poor developing economy into one of the richest economies in the world
- Even small differences in economic growth rates sustained over a number of years bring enormous differences in the standard of living

WHAT ARE THE PRECONDITIONS OF ECONOMIC GROWTH?

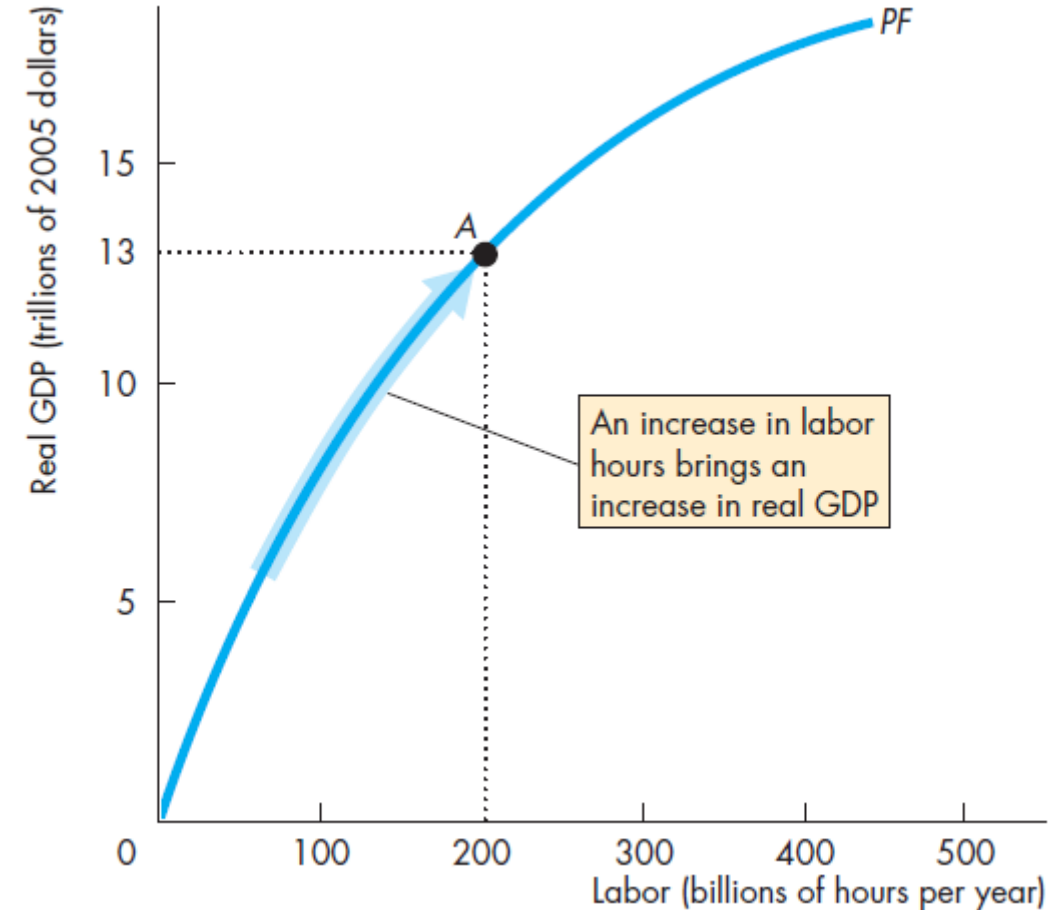


Potential GDP

- ***Reminder:** Potential GDP is the maximum level of real GDP that can be produced while avoiding shortages of labour, capital, land, and entrepreneurial ability that would bring rising inflation*
- On any given day, the quantities of land, entrepreneurial ability and capital are fixed, and they have a certain level of productivity which is given
- The quantity of labour employed is the only **variable factor** of production
- Hence, Potential GDP is the level of real GDP when the quantity of labour employed is the **full-employment quantity**
- Potential GDP can be determined by a model consisting of two components:
 - An aggregate production function
 - An aggregate labour market

Aggregate Production Function

- **Aggregate production function:** The aggregate production function is the relationship that tells us how real GDP changes as the quantity of labour changes when all other influences on production remain the same.
- An increase in the quantity of labour brings a movement along the production function and an increase in real GDP
- The Aggregate Production Function is labelled as *PF*.



Aggregate Labour Market

- ***Aggregate labour market:*** A market that determines the quantity of labour employed and the quantity of real GDP produced

Each additional hour of labour increases real GDP by successively smaller amounts (***Law of Diminishing Returns***)

- ***The Real Wage Rate:*** It is the money wage rate divided by the price level

The real wage rate is the quantity of goods and services that an hour of labour earns; hence, it is the price of labour

It influences both the supply of and the demand for labour

Aggregate Labour Market

- ***The Demand for Labour:*** It is the relationship between the quantity of labour demanded and the real wage rate.

The quantity of labour demanded is the number of labour hours hired by all the firms in the economy during a given period

The quantity of labour demanded ***increases*** as the real wage rate ***decreases***

The demand for labour curve is labelled *LD*

- ***The Supply of Labour:*** It is the relationship between the quantity of labour supplied and the real wage rate

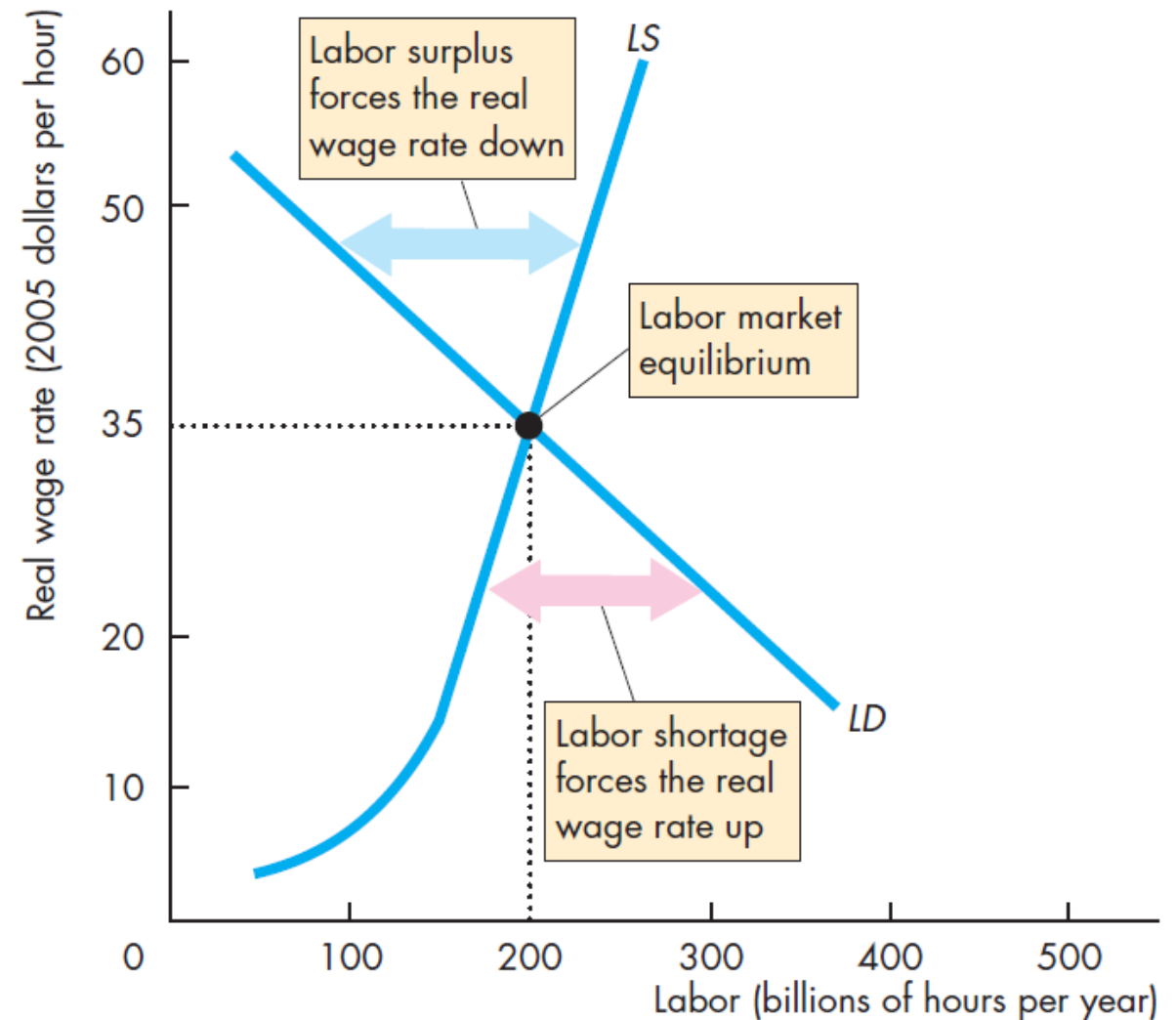
The quantity of labour supplied is the number of labour hours that all the households in the economy plan to work during a given period

The quantity of labour supplied ***increases*** as the real wage rate ***increases***

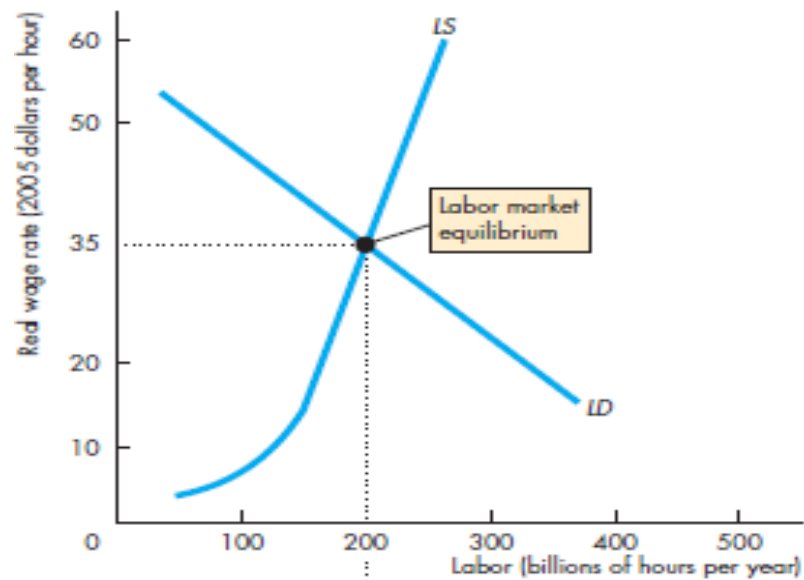
The supply of labour curve is labelled *LS*

Aggregate Labour Market

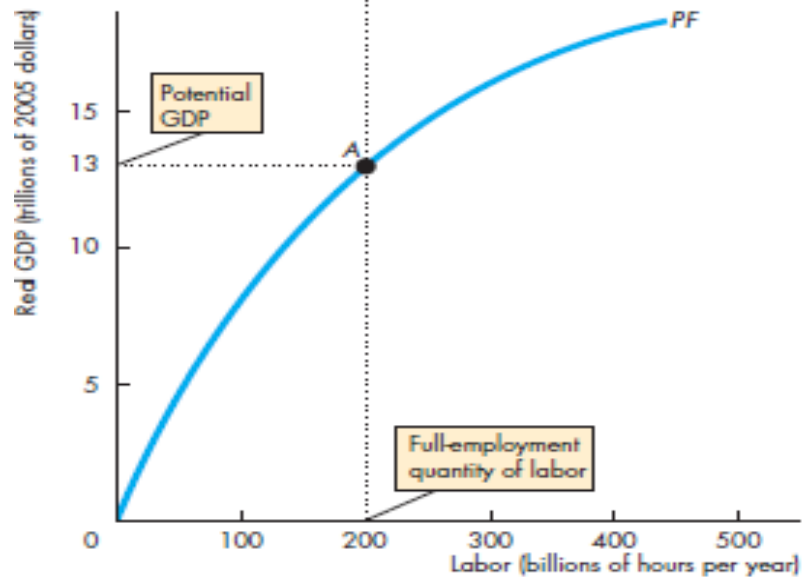
- Labour Market Equilibrium: When there is neither a shortage nor a surplus, the labour market is in equilibrium; a ***full-employment*** equilibrium.



At the equilibrium quantity of labour, the economy is at full employment, and the quantity of real GDP at full employment is potential GDP



(a) The labor market



(b) Potential GDP

Growth of Potential GDP: Supply of Labour

- ***Growth of the Supply of Labour:*** When the supply of labour grows, the LS curve shifts rightward; The quantity of labour at a given real wage rate increases
- The quantity of labour can change due to:
 - Average hours per worker:
 - The employment-to-population ratio
 - The working-age population
- Average hours per worker have decreased as the work-week has become shorter, and the employment-to-population ratio has increased as more women have entered the labor force; which has kept the average hours per working-age person (approximately) constant
- Hence, growth in the supply of labour has primarily been due to an increase in the working age population; which is caused by ***population growth***

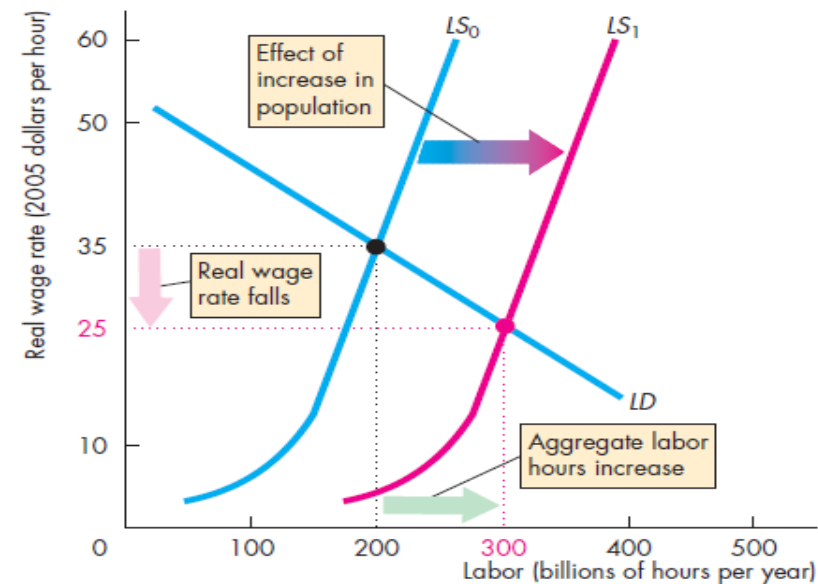
Growth of Potential GDP: Supply of Labour

- ***The Effects of Population Growth:*** Population growth brings growth in the supply of labour, but does not change the demand for labour or the production function
- With an increase in the supply of labour and no change in the demand for labour, the real wage rate falls and the equilibrium quantity of labour increases
- The increased quantity of labour produces more output, and potential GDP increases

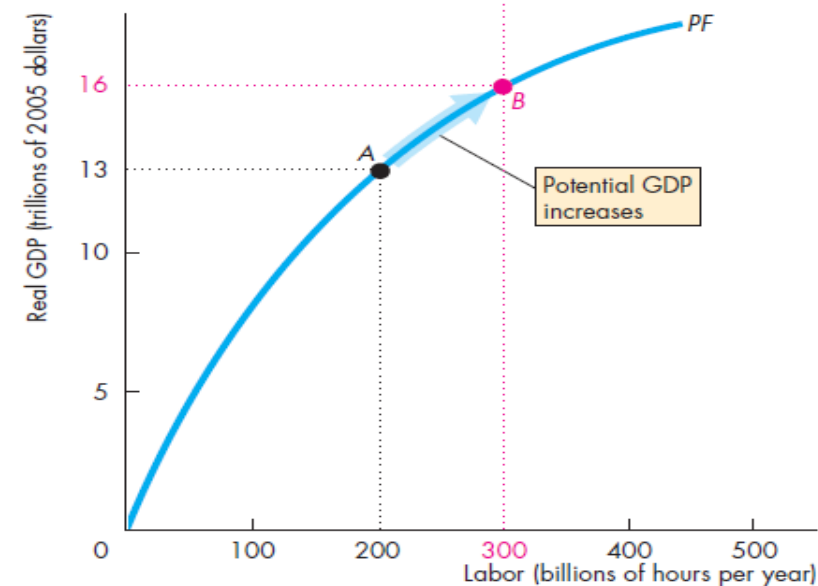
An increase in the population increases the supply of labour.

Causing the LS curve to shift rightward. The real wage rate falls and aggregate labour hours increase.

The increase in aggregate labour hours brings an increase in potential GDP. But diminishing returns bring a decrease in potential GDP per hour of labour



(a) The labor market



(b) Potential GDP

Some Key Review Questions

- What would happen if other things remained constant and only the average hours per worker/the employment-to-population ratio changed?
- Why are there diminishing returns to the increase in real GDP for each additional hour of labour increase?
- Think about some real-life scenarios that might cause the supply of labour to grow

Important Note!!!

***GO THROUGH THE FIGURES IN CHAPTER 23 FOR A BETTER
UNDERSTANDING OF THE SHIFTS OF THE CURVES!!!***